ADARSH ROUNIYAR

Software Engineering Leadership | Secure Systems Architecture | Production Delivery

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LinkedIn | Technical Portfolio | WStaking

PROFESSIONAL PROFILE

Software engineering technical lead with experience directing cross-functional product delivery, supervising 3–4 frontend developers, and owning architecture, secure integration, release quality and production operations. Led WStaking from coordinated design and implementation to full-stack operational ownership; the platform displayed US$2.079 million in total staked assets in October 2026. M.Sc. in Computer Science and Information Engineering with peer-reviewed research in trustworthy AI and smart-city systems.

CORE COMPETENCIES

Technical leadership: Developer supervision, task allocation, implementation reviews, technical acceptance, defect prioritization, stakeholder coordination, progress reporting.
Software engineering: Solution architecture, API and database design, legacy/integration workflows, TypeScript, React/Next.js, Node.js, Python, Solidity, PostgreSQL, Firestore.
Quality and operations: Automated and regression testing, code review, access control, audit remediation, incident response, deployment validation, rollback, monitoring, Linux/Docker.
AI and research: LLM infrastructure, retrieval-augmented generation, federated learning, TensorFlow/PyTorch, technical documentation and research communication.

PROFESSIONAL EXPERIENCE

AI Software Development Engineer | WStaking Project Technical Lead **Apr 2024 – Present**
Shixun Technology Co., Ltd. (employer); assigned to Whale Strategy Co., Ltd. | Taiwan

- Accountable for WStaking technical delivery across frontend, backend services, wallet integrations and smart contracts on BNB Chain, Base and X Layer; aligned design and implementation workstreams.
- Supervised 3–4 frontend developers with project-level authority to assign tasks, review implementations, test integrations, track defects and approve technical deliverables.
- Coordinated with a UI/UX designer and bilingual project manager to translate client requirements into Figma design handoffs, implementation specifications and integration acceptance criteria.
- Designed and maintained backend and data workflows for staking, rewards, referrals, wallet binding, claims, reconciliation and administrative operations using Node.js, PostgreSQL and Firestore.
- Implemented signed administrative actions, access controls and transaction validation; coordinated external smart-contract audit findings, remediation and release checks.
- Proposed a post-launch consolidation of frontend maintenance; management approved the restructuring, after which I assumed end-to-end frontend, backend and smart-contract maintenance.
- Investigated wallet connectivity, RPC outages, data consistency and referral accounting incidents; used automated tests, staged verification and rollback planning for production fixes.
- Established and operated internal AI/LLM engineering services; communicate technical priorities, delivery progress, incidents and risks directly to company leadership.

AI Research Assistant & Project Assistant **Nov 2021 – May 2024**
National Chung Cheng University (CCU) | Taiwan

- Researched trustworthy and federated AI for sustainable smart-city applications under work associated with an NSTC-supported research project; studied distributed learning, model robustness and environmental sensing.
- Developed and evaluated CNN-based air-pollution estimation models and distributed learning pipelines in TensorFlow/PyTorch; documented experiments, datasets, architecture and findings.
- Worked with academic and international research collaborators to present architectures, experimental results and reproducible research outputs.
- Co-authored peer-reviewed work published in IEEE ISC2, ACM GoodIT, *Future Internet* and *Artificial Intelligence Review*, covering trustworthy AI, federated learning and environmental monitoring.

SELECTED ENGINEERING & RESEARCH WORK

WStaking — Production multi-chain platform | wstaking.net
Led delivery and now operate a multi-chain platform across BNB Chain, X Layer and Base. Integrated staking, reward claims, renewals, full/partial unstaking and on-chain event processing; designed signed referral binding, reconciliation and auditable administrative settlement. Built Telegram-based monitoring and resolved wallet, RPC, data-consistency and reward-accounting incidents. Displayed total staked: US$2,079,185 (17 October 2026; not revenue).

Whale Strategy Internal AI Platform | Platform
Designed and deployed multi-user LLM infrastructure using Ollama, llama.cpp, Docker, Cloudflare Tunnel and OpenAI-compatible APIs; configured model routing and developer access through VS Code Continue. Maintained deployment and operational configuration.

Almond teAI — Code Security Scanner & Trust Platform | Live application
Developed a full-stack Next.js/TypeScript security analysis platform with Prisma and SQLite: static analysis, secret/dependency detection, security scoring, regression tracking, PDF reports and EVM smart-contract review workflows.

WS-AI Crypto Assistant | Live application
Implemented LLM and retrieval-augmented generation workflows with live market-data integration, contextual analysis, sentiment processing and decision-support outputs.

Automated Trading Infrastructure | Repository
Developed and operated Docker-based Python trading services with ML-assisted filtering, OKX API integration, monitoring and Linux process management.

Eff-AQI — Federated Learning for Air Pollution Estimation | ACM publication
Built a CNN-based air-pollution estimation research system and supporting image dataset; published findings at ACM GoodIT and released research resources through GitHub.

ADDITIONAL PRODUCTION ENGINEERING

Reliability and recovery: Recovered services after a primary NVMe failure through backup preservation, OS/environment restoration and staged service recovery. Built transaction and wallet-event monitoring, access-anomaly alerts and RPC/indexer fallback mechanisms.
Release assurance: Investigated production incidents using reproducible tests, verified data/chain evidence, controlled rollouts and rollback readiness; coordinated external smart-contract audit findings and remediation.

EDUCATION

M.Sc., Computer Science and Information Engineering 2021 – 2023
National Chung Cheng University, Taiwan GPA: 4.11/4.3
B.Tech., Computer Science and Engineering 2017 – 2021
Vel Tech University, India CGPA: 8.3/10

SELECTED PEER-REVIEWED PUBLICATIONS

- "Federated Learning Design and Functional Models: Survey." *Artificial Intelligence Review*, 2024. DOI
- "Federated Adversarial Training Strategies for Sustainable Smart Cities." *Future Internet*, 2023. DOI
- "Eff-AQI: Efficient CNN-Based Air Pollution Estimation." *ACM GoodIT*, 2023. DOI
- "Federated Trustworthy AI Architecture for Smart Cities." *IEEE ISC2*, 2022. DOI

LANGUAGES

Nepali, English, Hindi; basic Mandarin Chinese.